RecallGuard

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RecallGuard

Estimating Product Recall Risk from Consumer Safety Complaints

From messy public safety data to an ML-based risk-assessment system.

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MOTIVATION

Recalls often follow a paper trail of warning signs.

15,000+

product types fall under CPSC jurisdiction

Public complaint data may contain early warning signs of unsafe products, but the volume and inconsistency of the records make manual review impractical.

01 Recalls often happen after warning signs appear in the public record.

02 Consumer complaints contain early signals: hazard descriptions, repeated failures, injury reports.

03 The data is too large and too inconsistent to read manually.

04 A model could help triage products for human review earlier.

RESEARCH QUESTIONS

Three questions guided the project.

Q 1

Can complaint text help predict recall risk?

Treat consumer narratives and product descriptions as the primary signal.

Q 2

Can structured product details improve that prediction?

Combine text features with category, brand, and product-type information.

Q 3

Can this be packaged as a practical safety-assessment tool?

Move beyond a notebook into an application a non-technical user can interact with.

THE RAW DATA

Two public CPSC datasets, neither built for joining.

Source: U.S. Consumer Product Safety Commission (CPSC). Publicly available government data.

Incident Reports

from SaferProducts.gov

- Consumer-submitted complaint reports
- Product descriptions, brands, model numbers
- Free-text narratives of what happened
- Reviewed by CPSC staff before publication

Recalls Listings

from cpsc.gov/Recalls

- Official recall announcements
- Recalled product details and hazard descriptions
- Recall dates, product codes, UPCs
- CPSC has jurisdiction over 15,000+ product types

Why I had to connect them

Incident reports do not say which products were later recalled. Recall listings do not contain complaint-level training examples. **There is no shared key, so I had to build my own linkage.**

THE LINKAGE PROBLEM

Why a simple join wouldn't work.

No shared product ID

Neither file contains a universal key linking the two.

UPC often missing

The most reliable identifier was absent from most rows.

Inconsistent brand names

"GE", "General Electric", "GE Appliances", all the same brand.

Messy model numbers

Spaces, prefixes, and free-text formatting in both files.

Long, unstructured descriptions

Product names embedded in paragraphs of consumer prose.

A simple join returned almost nothing

Direct equality on any one field missed most true matches.

FIRST ATTEMPT

Fingerprint-Based Linking.

THE APPROACH

Treat product descriptions as fingerprints, then match.

1. Clean and normalize product descriptions.
2. Extract model-like tokens from the text.
3. Treat these tokens as product “fingerprints.”
4. Link incidents to recalls using fingerprint overlap.
5. Build the first recalled vs. not recalled training labels.

WHAT'S A MODEL-LIKE TOKEN?

A short string that looks like a product model or code, embedded in messy text.

EXAMPLES FROM THE DATA

JS645SL3SS	LRE30755ST
BEC-1500	A1881

More useful than generic words like “oven” or “charger.”

THE OUTCOME

Severe class imbalance and weak results. Time to revisit the linkage itself.

WHY IT HAD TO BE REBUILT

Why the Fingerprint-Only Dataset Had to Be Rebuilt.

Fingerprint overlap helped

It was enough to find candidate matches between the two files.

Some matches were still weak

Loose overlaps pulled in low-confidence links and labeled them as positive.

"Not matched" ≠ true negative

An unmatched row only means we couldn't link it. It does not mean the product was safe.

Ambiguous rows created noise

Edge cases got mixed straight into the training set instead of being held out.

A more selective rebuild was needed

Keep the linkage idea. Tighten the matching. Filter the labels.

THE TAKEAWAY

The bottleneck wasn't the model. It was the labels.

Before trying more algorithms or rebalancing the data, the linkage approach itself needed to be tightened.

REBUILDING THE TRAINING DATA

Four steps that turned noisy labels into clean ones.

1 NORMALIZE

- Standardize brand, manufacturer, model, UPC, product code, category, sub-category, type.
- Lowercase, clean, strip formatting noise.
- Make values comparable across both datasets.

2 MATCH

- Exact model + brand
- Exact UPC
- Exact product code
- Token overlap (fingerprints)
- Category agreement
- Fuzzy similarity

3 ASSIGN CONFIDENCE

- HIGH: exact UPC, product code, or model + brand
- MEDIUM: meaningful but partial evidence
- AMBIGUOUS: weak or conflicting overlap
- NONE: no recall evidence

4 LABEL

- Keep HIGH-confidence positives.
- Do NOT force AMBIGUOUS into negatives.
- Sample clean negatives from NONE.
- Save training_data.csv.

The output, training data file, is what the ML models actually learn from. The models do not learn directly from the raw recall file.

FINAL TRAINING DATASET

What the linkage pipeline produced.

training_data.csv · the supervised dataset every downstream model learns from

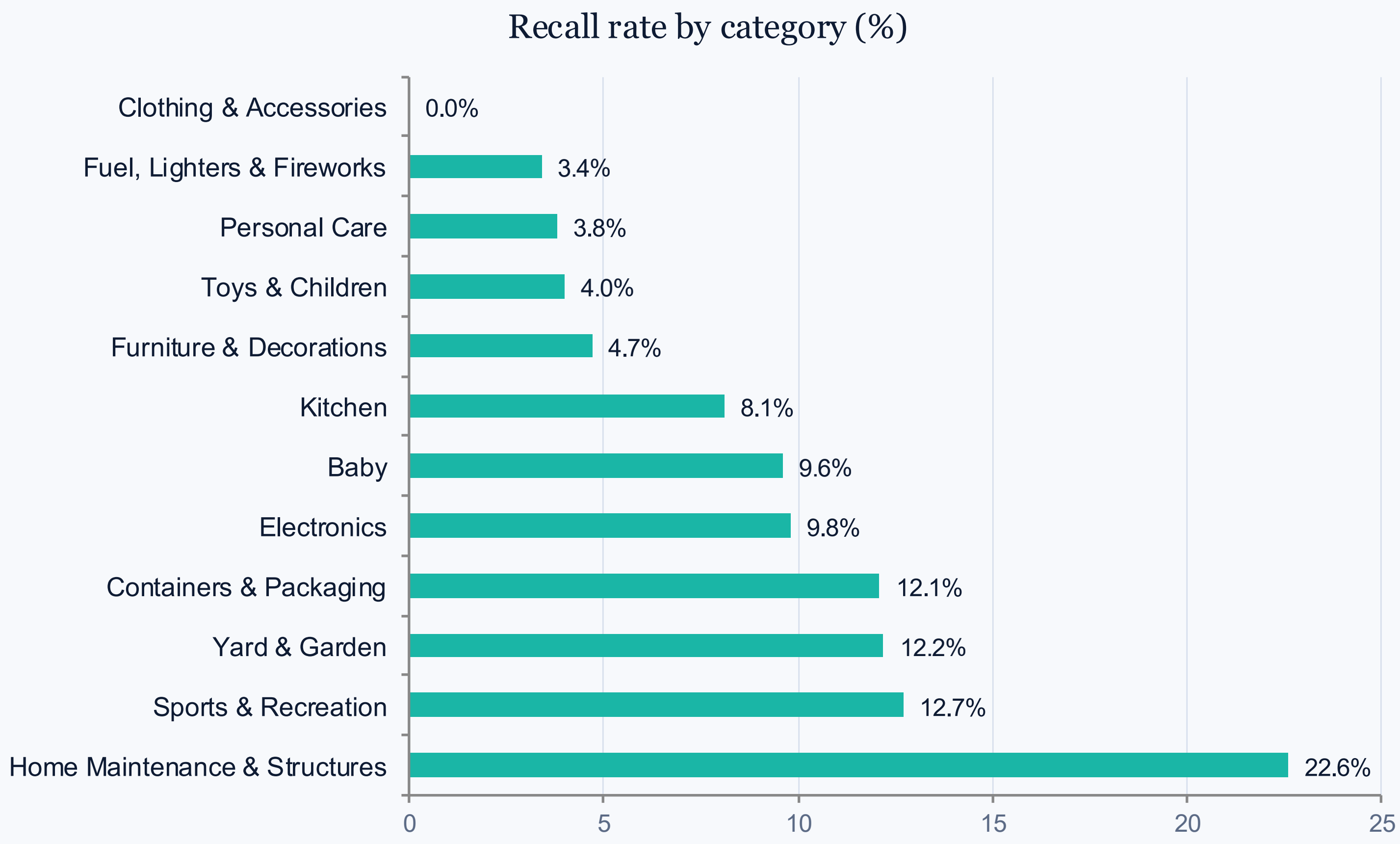
3,780	64	420	3,360	11.1%
Total incident reports	Columns	Recalled	Not recalled	Positive rate

Strong class imbalance. Only ~1 in 9 incidents is linked to a recall.

Imbalance shapes every modeling decision that follows.

EXPLORATORY ANALYSIS

Recall rates differ sharply by product category.



Home Maintenance leads

22.6% recall rate. Far above the 11.1% baseline.

Long tail of low-rate categories

Most categories sit between 4 and 10%.

Class imbalance is global

3,360 not recalled vs. 420 recalled across the full dataset.

Sparse fields throughout

Many columns are 60 to 97% missing. Limits direct use as features.

TEXT ANALYSIS

Length wasn't enough, but the vocabulary was rich.

WHAT I CHECKED

Length alone is not a signal

- Combined product description and incident narrative into one text field.
- Cleaned, lower-cased, removed stop words.
- Compared word-count distributions across recalled vs. not-recalled cases.
- **Recalled and non-recalled narratives have similar average length.**

TOP NON-STOP TERMS

What the model would actually see

consumer	kitchen	model	oven
home	samsung	range	glass
unit	door	maintenance	electric
structures	plastic		

Strong product and hazard vocabulary. Exactly what TF-IDF can exploit.

FEATURE ENGINEERING

Translating raw fields into model-ready signals.

TF-IDF on product description	Word weights that emphasize distinctive product terms.
TF-IDF on incident narrative	Same technique on consumer text. Keeps hazard vocabulary separate.
Character n-grams	Captures partial brand and model strings whole-word features miss.
One-hot encoding	Category, sub-category, and product type as 0/1 columns.
Extra numeric features	Word counts, hazard-word counts, has-model-info flag.

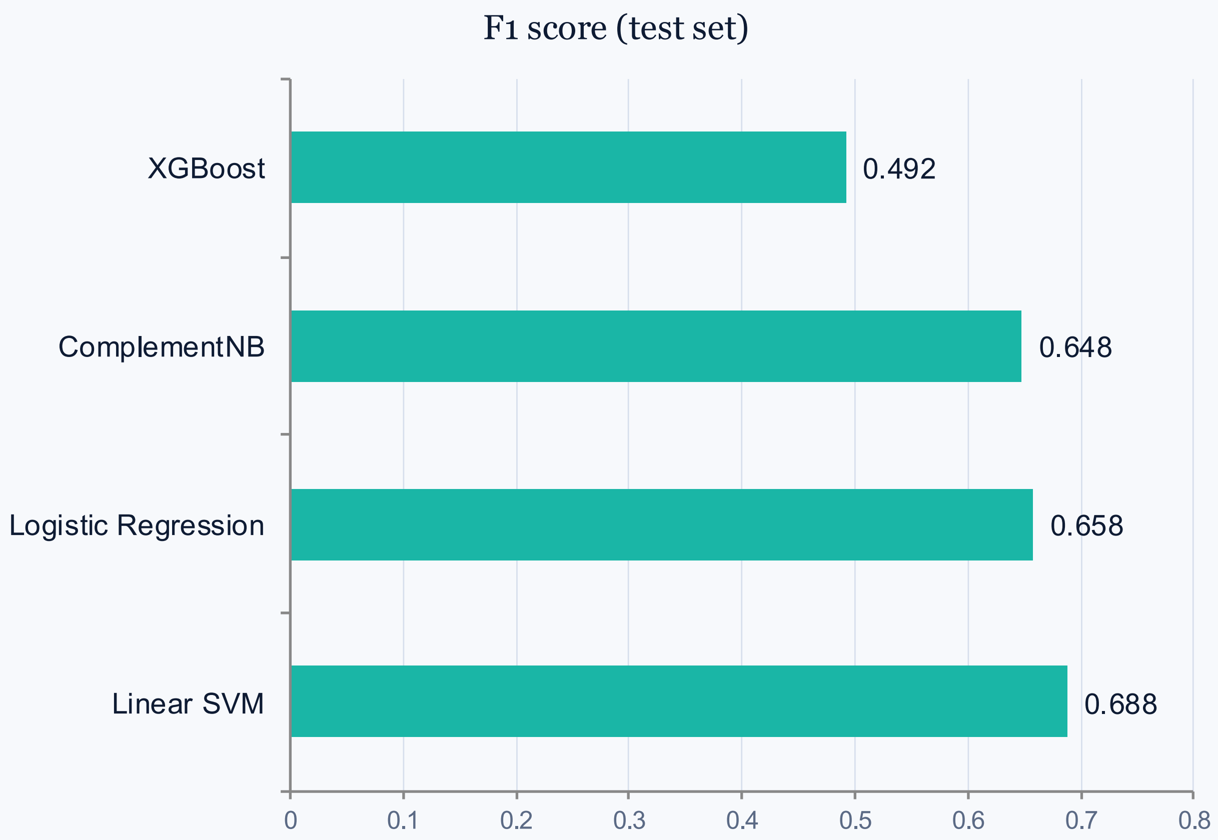
MODELS CONSIDERED

Four classical algorithms, plus two imbalance fixes.

MODEL	WHAT IT DOES	WHY I USED IT
Logistic Regression	Linear classifier; outputs a probability.	*Strong text baseline; gives calibrated probabilities.*
Linear SVM	Finds the best linear separator between classes.	*Strong on sparse text features.*
Complement Naive Bayes	Bayesian classifier adapted for imbalanced text.	*Built for this kind of imbalance.*
XGBoost	Gradient-boosted trees; non-linear.	*Industry benchmark for tabular data.*
Threshold tuning	Adjusts the probability cutoff for predictions.	*Trades precision against recall.*
SMOTE	Synthesizes new positive examples.	*Standard remedy for class imbalance.*

GENERAL MODEL COMPARISON

Best-performing model: Linear SVM. XGBoost lagged.



MODEL	ACC	PREC	RECALL	F1
Linear SVM	**0.934**	**0.724**	**0.655**	**0.688**
Logistic Regression	0.933	0.754	0.583	0.658
ComplementNB	0.933	0.770	0.560	0.648
XGBoost	0.918	0.789	0.357	0.492

KEY TAKEAWAYS

Linear SVM leads on F1 (0.688).

Logistic Regression is close, and gives probabilities for the app.

ComplementNB is competitive but slightly weaker.

XGBoost underperforms. High precision, low recall.

F1 = balance of precision and recall (0 to 1). Better than accuracy on imbalanced data.

THRESHOLD TUNING & SMOTE

Two data imbalance remedies. Neither beat the SVM.

THRESHOLD TUNING

Adjusting the probability cutoff

Lowering the 0.5 cutoff catches more true recalls but flags more false alarms. Swept 0.10 to 0.90 on Logistic Regression.

SMOTE

Synthesizing minority-class examples

Synthesizes new positive examples by interpolating between existing ones. Rebalances training without dropping data. Applied to LR and SVM.

RESULT

Both experiments helped some models slightly, but **neither beat the un-tuned Linear SVM on F1.**
The lesson: better data and better features mattered more than algorithmic tricks for imbalance.

CATEGORY-SPECIFIC MODELING

Why a focused model can outperform a broad one.

RATIONALE

One model per category. Same pipeline, narrower scope.

- Different product groups, different hazard vocabularies.
- A broad model averages across them; a focused model learns the signal cleanly.
- Chose Home Maintenance & Structures: enough rows, highest recall rate (22.6%).
- Same pipeline, applied to that category only.

BEST CATEGORY-SPECIFIC RESULT

Home Maintenance & Structures

Accuracy	Precision	Recall	F1
0.899	0.763	0.806	0.784

F1: 0.688 → 0.784 vs. the general model.

FROM CLASSIFIER TO RISK SYSTEM

Three signals, one composite score.

ML MODEL PROBABILITY

Model-estimated likelihood of recall (0 to 1).

from Logistic Regression

HAZARD SEVERITY SIGNAL

Reads incident text for severity cues like fire, injury, burn, or choking.

from text rules

DETECTED RISK TERMS

Surfaced keywords so the score is never a black box.

from category-specific keywords

COMPOSITE RISK SCORE

ML probability + Hazard severity

A single percentage with a Low / Medium / High band and a suggested action.

PIPELINE OVERVIEW

Putting it all together: the full system, end to end.

STAGE 1	STAGE 2	STAGE 3	STAGE 4	STAGE 5
RAW DATA	DATA PREPROCESSING	FEATURE ENGINEERING	ML DEVELOPMENT & ANALYSIS	STREAMLIT TOOL
Incident Reports + Recalls Listings	Linkage, labeling, sampling	TF-IDF · n-grams · one-hot encoding	LR · SVM · CNB · XGBoost	Composite risk score
Two unjoined CPSC datasets	3,780 rows · 420 recalled	Text + structured + numeric	Best model: Linear SVM	Risk explorer · scorer · model details

Five stages, one system. Next: the Streamlit app that wraps it for end users.

FUTURE WORK

Five directions to extend RecallGuard.

1. **Product URL input**
   Paste a URL instead of typing fields by hand.
2. **External review aggregation**
   Pull retailer reviews to score products not in the CPSC data.
3. **Score products outside the training set**
   Lean on text features rather than category lookups.
4. **LLM interpretation layer**
   Plain-English explanations of why a product was flagged.

CONCLUSION

Five things to take away.

1. The hardest part was turning two unjoined datasets into a usable training set.
2. Rebuilding the data lifted F1 from mid 0.3 to 0.69. Better data beat any modeling change.
3. Linear SVM was the best-performing general model (F1 = 0.688).
4. Category-specific modeling pushed F1 to 0.784 on Home Maintenance & Structures.
5. RecallGuard is a transparent composite risk-assessment system, not just a classifier.

Thank you.

Questions welcome.